

Hyungju Lee

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SUMMARY

Product data scientist and analytics engineer who owns the measurement layer of a subscription mobile and streaming product end to end: source-of-truth KPI definitions, event instrumentation, the SQL and Python pipelines underneath them, and the reporting executives make decisions from. Currently responsible for product analytics across a 138-client portfolio with six years of longitudinal history, covering retention, churn, engagement, conversion, and subscription revenue. Statistics-trained, with an M.E.Sc and peer-reviewed machine learning research in LiDAR 3D object detection.

CORE SKILLS

- **Languages & Data:** SQL, Python, R, PostgreSQL, Athena, Pandas, Polars, Parquet, Excel
- **Product Analytics & Measurement:** KPI design, source-of-truth metrics, DAU/WAU/MAU, retention and churn, engagement scoring, conversion, membership lifetime, subscription and revenue analytics, cohort-style reporting
- **Analytics Engineering:** ETL and batch reporting pipelines, Parquet-backed KPI datasets, scheduled jobs, reporting automation, data quality, reconciliation, dashboards, AWS, Docker
- **Statistics & Measurement Design:** Experimental design, statistical testing, quasi-experimental pre/post impact analysis, quantitative evaluation, decision support
- **Instrumentation & Delivery:** First-party event and clickstream instrumentation, tracking audits, Dash, Plotly, stakeholder and executive communication

EXPERIENCE

MySeat Media | Product Data Scientist, Analytics Engineering & Monetization | Toronto, Canada

2023 - Present

- Built and standardized source-of-truth KPI frameworks across engagement, retention, churn, memberships, livestream activity, and revenue for a 138-client mobile and streaming portfolio.
- Built SQL and Python analytics pipelines and longitudinal datasets spanning 6.65M+ sessions, transformed into Parquet-backed KPI datasets powering dashboards, benchmarking, and recurring reviews.
- Designed a cross-platform subscription and revenue model across Apple IAP, Google Play, and Stripe/PWA, reconciling 19.6K+ records to make billing and monetization reporting trustworthy.
- Instrumented first-party event and clickstream pipelines across devices, sessions, user actions, music activity, livestream engagement, and push interactions, spanning mobile clients and backend analytics services.
- Designed and analyzed quasi-experimental pre/post studies on push notifications, livestreams, content changes, and auction changes, measuring effects on engagement, conversion, and revenue.
- Automated recurring executive and client reporting through 34 scheduled jobs and precomputed dashboard refresh workflows using PostgreSQL, Athena, Pandas, and Mixpanel exports.
- Built stakeholder dashboards for conversion, churn, engagement, geo-distribution, and monetization trends with role-based client access, giving executives and cross-functional partners self-serve visibility.
- Translated ambiguous requests from finance, product, leadership, engineering, and client-facing teams into scoped analyses, trusted metrics, and decision-ready recommendations.

AUSM Lab | Machine Learning Researcher | Toronto, Canada

2019 - 2024

- Designed, ran, and evaluated PyTorch deep learning experiments for LiDAR point-cloud 3D object detection, owning experiment configuration, benchmarking, and quantitative evaluation against public and custom datasets.
- Published peer-reviewed research at ICPR 2022 and ISPRS 2023 and built reproducible annotation and evaluation workflows producing 5,000+ labeled objects.

Ontario Ministry of Natural Resources and Forestry | Financial Analyst | Peterborough, Canada

2017 - 2018

- Built and automated Oracle-based financial reporting workflows used by project managers and senior leadership across ministry divisions.

EDUCATION

- M.E.Sc., Earth and Space Science & Engineering, York University (2020 - 2022) | Best Master's Thesis Award (2022)
- H.B.Sc., Statistics, Quantitative Finance Stream (Co-op), University of Toronto (2015 - 2020)

PUBLICATIONS

- *YUTO Tree-5000: A Large-scale Airborne LiDAR Data for Single Tree Detection* — ICPR 2022, Montreal
- *YUTO: A Large Scale Aerial LiDAR Data Set for Semantic Segmentation* — ISPRS 2023, Cairo